KAUST

AMCS 206 - Applied Numerical Methods

Midterm - March 27, 2023

Duration: 80 minutes.
Clearly justify all your answers and all the steps taken to explain your reasoning.

Problem 1 [40 points]

The Taylor expansion of $\ln(1+x)$ about 0 is

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots = \sum_{k=1}^{+\infty} (-1)^{k-1} \frac{x^k}{k}, \quad -1 < x \leq 1.$$

1. a) [5 points] Determine the Taylor expansion of order 5 of $f(x) = \ln x$ about $x_0 = 1$ using the Taylor expansion of $\ln(1+x)$ about 0. Show your work.

   b) [10 points] Let $x > 1$. Give the analytical expression for the error of the Taylor expansion of order 5 of $f(x) = \ln x$, as well as an upper bound for this error. Show your work.

   c) [10 points] Suppose we want to approximate $\ln 2$ using the Taylor expansion of $f(x) = \ln x$ about $x_0 = 1$. How many terms are required in the expansion to have an absolute error less than 10^{-6}? Is this a good method to compute $\ln 2$? Justify.

2. a) [5 points] Find the Taylor expansion of

$$\ln\left(\frac{1+x}{1-x}\right),$$

   about $x_0 = 0$. Show your work. Hint: use the Taylor expansion of $\ln(1-x)$.

   b) [10 points] Do you expect this Taylor expansion to be a better method to compute $\ln 2$? Discuss.

PROBLEM 2 [40 POINTS]

Consider the following definition and results:

- An $n \times n$ matrix $\mathbf{C}$ is strictly diagonally dominant if and only if

$$|c_{ii}| > \sum_{j=1, j\neq i}^{n} |c_{ij}|$$

  holds for each $i = 1, 2, \ldots, n$.

- If $\mathbf{C}$ is a strictly diagonally dominant matrix, then for any choice of $\mathbf{x}^{(0)}$, both the Jacobi and the Gauss-Seidel methods give sequences that converge to the unique solution of $\mathbf{Cx} = \mathbf{b}$.

- If $\mathbf{C}$ is a strictly diagonally dominant matrix, Gaussian elimination can be performed on any system $\mathbf{Cx} = \mathbf{b}$ to obtain its unique solution without row or column interchanges.

Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} \alpha & 0 & \gamma \\ 0 & \alpha & \beta \\ 0 & \delta & \alpha \end{bmatrix},$$

in which α, β, γ and δ are real parameters. We want to solve $\mathbf{Ax} = \mathbf{b}$.

1. [5 points] Give a set of sufficient conditions on α, β, γ and δ to have convergence of the Jacobi and the Gauss-Seidel methods for $\mathbf{A}$.

2. [10 points] Assume $|\alpha| > |\gamma|$ and $|\alpha| > |\beta|$ and $|\alpha| > |\delta|$ and perform the $(\mathbf{P^T L})\mathbf{U}$ factorization of $\mathbf{A}$.

3. Both the Jacobi and the Gauss-Seidel methods are stationary iterative methods that can be written as

$$\mathbf{x^{(k+1)}} = \mathbf{x^{(k)}} + \mathbf{M^{-1}r^{(k)}}.$$

   a) [5 points] What is $\mathbf{r^{(k)}}$?

   b) [5 points] Let $\mathbf{M_J}$ be $\mathbf{M}$ for the Jacobi method and $\mathbf{M_{GS}}$ be $\mathbf{M}$ for the Gauss-Seidel method. What are $\mathbf{M_J}$ and $\mathbf{M_{GS}}$ for the given $\mathbf{A}$?

   c) [5 points] Do we ever need to calculate $\mathbf{M^{-1}}$? Explain.

4. If $\mathbf{e^{(k)}} = \mathbf{x} - \mathbf{x^{(k)}}$ is the error at iteration $\mathbf{k}$, then

$$\mathbf{e^{(k+1)}} = (\mathbf{I} - \mathbf{M^{-1}A})\mathbf{e^{(k)}}.$$

   Assume $|\alpha| > |\gamma|$ and $|\alpha| > |\beta|$ and $|\alpha| > |\delta|$.

   a) [5 points] Is it true that

$$\rho\left(\mathbf{I} - \mathbf{M_J^{-1}A}\right) < 1?$$

   Why or why not?

   b) [5 points] Is it true that

$$\rho\left(\mathbf{I} - \mathbf{M_J^{-1}A}\right) < \rho\left(\mathbf{I} - \mathbf{M_{GS}^{-1}A}\right)?$$

   Why or why not?

Problem 3 [20 points]

Suppose you need to fit data (x_i, y_i), with $i = 1, \ldots n$ and $x_i > 0, y_i > 0$, to the power function

$$y = bx^m$$

using the linear least squares method. Take the natural logarithm on both sides of $y = bx^m$ and explain how to determine the best fitting values for m and b, in the least squares sense.